

Throwing bits at K4: Evidence for a Three-Zone Split with Tableau-Keyed XOR Core

DRAFT

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Abstract

This paper presents a theory that K4, the unsolved 97-letter passage of *Kryptos*, is split into three zones, enciphered with different, most likely two, methods. A 61-letter core body (positions 25–85) is enciphered by a five-bit XOR against the sculpture’s own engraved Vigenère tableau, read from the start of row M and aligned so that the row’s first letter falls immediately after the opening four-letter group OBKR, whose first two letters. The head (positions 0–24) and tail (86–96) flanks are enciphered by a substitution method, with no XOR or use of the tableau. We present the statistical evidence for these conclusions and show an example of how the known BERLINCLOCK crib would have been encrypted. The paper makes no claim of recovering additional plaintext of K4 beyond what is publicly known.

1 Introduction

K4 is the 97-character ciphertext

OBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

with two artist-disclosed plaintext fragments (*cribs*): EASTNORTHEAST at positions 21–33 and BERLINCLOCK at 63–73 (0-based throughout). The K4 ciphertext has resisted all known decryption attempts, no plaintext has been decrypted and no method of K4’s encryption has been conclusively shown.

In this paper we perform computer-assisted searches and statistical analysis to show that K4 has a distinct 61-character *body* and *flanks*, with the longer body segment showing interaction well beyond chance with the KRYPTOS-keyed Vigenère tableau engraved on the right hand half of the sculpture. The K4 architecture this paper proposes is summarised as:

The architecture this paper defends is shown in Figure 1 and summarised as:

1. **Three zones.** A head (0–24), a body (25–85), and a tail (86–96). The plaintext runs continuously across the seams — EAST ends the head and NORTHEAST opens the body — but the cipher changes at the boundary.

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2. **The body is XOR-encrypted against the sculpture.** Each body plaintext letter is first put through an unidentified substitution stage (the “mask”) and then combined, as a five-bit number, with the corresponding letter of the engraved Vigenère tableau, read linearly as carved from the first letter of row M’s content.
3. **The opening letters select the key.** The first two letters of K4 are 0B and they hint at row M as $O \oplus B = \text{‘M’}$. The sculpture’s DYHR raised-letter anomaly is a second, independent hint encoding M. The running key’s *alignment* is anchored at the end of the four-letter field — position 4 faces row M’s first letter, position 5 its second, and so on — though the XOR engages only inside the body. 0BKR itself is enciphered by the head’s substitution. Only 0 and B carry information.
4. **The flanks are a substitution.** The flanks - head and tail - are enciphered using a different method that does not involve the tableau or XOR operations. Both flanks are likely using the same method.

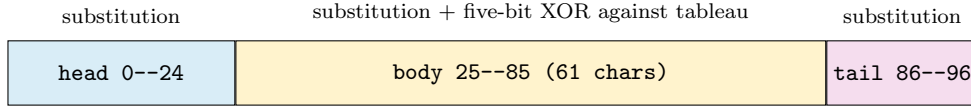


Figure 1: The architecture.

This paper does *not* provide a full decryption or any new plaintext. We make some brief arguments about decryption possibilities but they are of low confidence. In fact, if the paper’s overall architecture of K4 is correct, that is an argument against the plaintext being recoverable, as we will show.

AI Disclaimer: The analysis was performed, and this paper written, by the author who identifies as human. An LLM (Claude Opus) was used to assist with L^AT_EX formatting of tables and TikZ diagrams in this paper, as well as implement the ABC analysis in 6.1.

2 Methodology

Table 1 defines the terms we use in this paper.

Of particular importance to our method was the *shuffle null*, informally answering the question “would this finding be impressive if the ciphertext were meaningless?”. The search space for algorithmic approaches is large enough that algorithms will inevitably produce tempting looking partial matches. Examples the authors have seen include MSGFORCIA and BERLINCLOCKSPOOKSHAD. The infamous “K4 syndrome” is a tendency for an attempted solver of K4 to develop increasingly convoluted methods that go against the evidence and certainly seeing seemingly legible decryptions is a risk for going down that path. As such, we used shuffle nulls to protect ourselves during all algorithmic runs. We take the K4 ciphertext, scramble its letters into a random order, destroying any positional structure, and run the analysis on the scrambled copy. Doing this with a large number of scrambled texts lets us properly judge the significance of search results obtained when run against the real K4.

We index ciphertext positions as 0-based, 0–96. *Panel positions* index the 867-character engraved tableau.

2.1 Axioms

We treat every concrete clue provided by Jim Sanborn, the artist, as definitive. The cribs are assumed to be placed where Sanborn stated, and to be spelled correctly. The plaintext

Table 1: Terms used in this paper.

Term	Meaning
IoC	Index of coincidence:. Random uniform text ≈ 0.0385 ; English ≈ 0.0667 .
A5 encoding	Letters as five-bit numbers by alphabet position: A = 1, B = 2, ..., Z = 26.
XOR ($\oplus$)	Bitwise exclusive-or of two five-bit values.
passthrough	The convention for XOR results outside 1..26 (six of the 32 five-bit values have no letter): the input letter is emitted unchanged.
stream / intermediate	The result of undoing the XOR. This would have been the output of the substitution step, before applying the XOR, if our K4 architecture is correct.
mask	That substitution stage: the unknown plain $\rightarrow$ stream letter map.
indicator	Part of an encrypted message that configures the system, providing information about the key.
shuffle null	The control described above: the analysis re-run on letter-scrambled ciphertext. We derive p-values from these controls, e.g. “ $p = 0.002$ ” means 2 in 1000 scrambles did as well.
z-score	The standard score - how many deviations a value sits above the null’s mean.
total-variation distance	A 0-to-1 measure of how different two letter distributions are (half the summed absolute frequency differences).
keyed alphabet	An alphabet reordered by a keyword (KRYPTOSABCDE...). This alphabet, engraved with repeated shifts, is the tableau found on the sculpture..

is assumed to be exactly 97 characters long. These were implemented as hard constraints in every algorithmic search we ran.

In accordance with Sanborn’s statements, we rule out any method that requires knowledge of the K1-K3 plaintext. As such we don’t consider using the coordinates from K2 or any other part of the first three messages.

We also preemptively rule out any methods that are based on the particulars of the World Clock in Berlin. The first crib release happened twenty years after the sculpture was completed, it stands to reason that the message was intended to be decrypted without knowing it refers to the clock.

We similarly rule out any method requiring external information, such as book ciphers, or access to the actual sculpture. Our work assumes that an accurate transcript of the sculpture is enough to solve K4.

2.2 Initial assumptions

Our investigation was, before arriving at the results, guided by a number of clues and assumptions. These all appear reasonable to us but cannot be proven definitively. We list the clues and our assumptions below.

- Ed Scheidt asked “How do you deliver a message in such a way that it conveys not only the encrypted data, but also the key?”. Jim Sanborn has stated “I used a bit of stego [steganography] when designing the fourth part of Kryptos.”.
- The sculpture has a much-discussed irregularity in the engraving of the K3 ciphertext. A five-letter block **DYHR** appears visually distinct, with the letters Y, A, R in it elevated above the baseline. We assume this is significant and a strong candidate for a steganographic hint in the sculpture.
- Sanborn has multiple time referenced that he was taught “modern” encryption systems. We consider it likely he would have been taught at least the basics of digital encryption,

such as the principles behind DES. As bitwise XOR is a fundamental operation in many digital encryption systems, and is easily doable (and teachable) on paper, we consider it near certain that Sanborn was familiar with XOR.

Overall, our starting hypothesis was that, while K1-K3 are purely classical ciphers, K4 is a classical/digital hybrid, with a classical layer and a digital layer. The digital layer is assumed to be a bitwise operation, and it is assumed that the sculpture itself provides some steganographic hint about K4.

2.3 Hörenberg’s work

Cryptanalyst Michael Hörenberg has previously made observations in [1] that inform our digital approach to K4. Hörenberg assumed that the DYAHR anomaly was to be interpreted as a five-bit pattern, an assumption we do not rely on a priori but that our work corroborates. Hörenberg also used the tableau as a key stream source, which our work also does.

Several assumptions put forward in [1] have been definitely shown to be incorrect, as they contradict later information about K4, but our method independently arrives at the same running key as this earlier work.

3 Hiding in plain sight

3.1 The tableau as a key stream

If XOR encryption was used for a 97-letter text, a key of at least that length is needed. Examining the sculpture provides a candidate for the source of a key stream - the tableau panel. The sculpture’s second panel carries a Vigenère tableau keyed by KRYPTOS. Figure 2 reproduces the whole side as engraved. Read top left to bottom right, it is a single linear stream of letters: a 30-character header row (ABCD...XYZABCD), then 26 lettered rows of 31 characters each (row-label plus 30 content characters), and the header row is repeated again at the bottom. An unexplained quirk is that the N row has an extra character, a second L, being 32 characters long. In total,

$$30 + 26 \times 31 + 1 + 30 = 867 \text{ letters.}$$

A K4 candidate key is a 97-letter run of this stream, so candidates are determined by their starting position d on the panel. Disregarding the possibility of wrapped keys, the key readout has to fit inside the stream so d can be 0 through $867 - 97 = 770$, giving 771 candidate keys.

Alternatively, the key could be taken from the ciphertext panel as well, so a 97-long stretch of ciphertext preceding K4. This did not seem intuitive to us but we did run the ciphertext-as-key scan as control as well, as will be shown later.

3.2 The key search

Under this hypothesis, the first experiment is to evaluate each of the 771 possible tableau keys. For every offset d , take the 97-long key running from d to $d + 97$, XOR the K4 ciphertext with that key and measure the IoC to see if a non-random structure appears.

To perform the XOR, there are two questions that must be answered first.

1. What numerical values to assign to letters?
2. What rule to use for positions where the result of XOR is not a letter?

For numerical representation of letters, we tried several intuitive encodings.

ABCDEFGHIJKLMNQRSTUWXYZABCD	0
AKRYPTOSABCDEFGHIJLMNQUVWXZKRYPT	30
BRYPTOSABCDEFGHIJLMNQUVWXZKRYPT	61
CYPTOSABCDEFGHIJLMNQUVWXZKRYPTO	92
DPTOSABCDEFGHIJLMNQUVWXZKRYPTOS	123
ETOSABCDEFGHIJLMNQUVWXZKRYPTOSA	154
FOSABCDEFGHIJLMNQUVWXZKRYPTOSAB	185
GSABCDEFGHIJLMNQUVWXZKRYPTOSABC	216
HABCDEFGHIJLMNQUVWXZKRYPTOSABCD	247
IBABCDEFGHIJLMNQUVWXZKRYPTOSABCDE	278
JCABCDEFGHIJLMNQUVWXZKRYPTOSABCDEF	309
KDEFGHIJLMNQUVWXZKRYPTOSABCDEF	340
LEFGHIJLMNQUVWXZKRYPTOSABCDEF	371
MFGHIJLMNQUVWXZKRYPTOSABCDEFGHI	402
NGHIJLMNQUVWXZKRYPTOSABCDEFGHIJL	433
OHIJLMNQUVWXZKRYPTOSABCDEFGHIJL	465
PIJLMNQUVWXZKRYPTOSABCDEFGHIJLM	496
QJLMNQUVWXZKRYPTOSABCDEFGHIJLMN	527
RLMNQUVWXZKRYPTOSABCDEFGHIJLMNQ	558
SMNQUVWXZKRYPTOSABCDEFGHIJLMNQU	589
TNQUVWXZKRYPTOSABCDEFGHIJLMNQUV	620
UQUVWXZKRYPTOSABCDEFGHIJLMNQUVW	651
VUVWXZKRYPTOSABCDEFGHIJLMNQUVWX	682
WVWXZKRYPTOSABCDEFGHIJLMNQUVWXZ	713
XWXZKRYPTOSABCDEFGHIJLMNQUVWXZK	744
YXZKRYPTOSABCDEFGHIJLMNQUVWXZKR	775
ZZKRYPTOSABCDEFGHIJLMNQUVWXZKRY	806
ABCDEFGHIJKLMNQRSTUWXYZABCD	837

Figure 2: The tableau side of the sculpture: a header row, then 26 keyed rows whose first letter is the row label, then the header repeated. Row M, which will be of particular importance, is highlighted. Note the extra L at the end of the N-labeled row. The numbers on the right hand side show the panel index of the first letter in that row.

- A=1, B=2, ..., Z=26. Every letter fits within five bits. We call this encoding A5.
- Reversed A5, with Z=1 and A=26.
- A=0, B=1, ..., Z=25. We call this A0.
- KRYPTOS-alphabet indexed, K=1, R=2, ..., S = 7, A = 8, Z = 26.
- Baudot / ITA2 encoding. Included for historical interest and completeness despite being a legacy system by the time Kryptos was designed.

After applying the XOR to cipher and key, $C_i \oplus K_i$, if the result r was outside the range that letters are represented by, we tried several rules.

- Passthrough. Output the ciphertext letter C_i , unmodified.
- Output the keystream letter K_i .
- Modulo 26, so 27 becomes A, 28 becomes B, etc. $0 \rightarrow Z$.
- Truncate to the four lowest bits. Values 27 — 31 become K, L, M, N, O. $0 \rightarrow Z$.
- Reflect the result, so $52 - r$. 27 is same as 25 (Y), 28 is as 24 (X). $0 \rightarrow Z$.

Some of those choices like the inclusion of Baudot encoding or the “reflect” style rule appeared unlikely but the computational cost of these scans is effectively zero so we erred on the side of checking more.

Algorithm 1: tableau offset scan, with its calibrated null

```

OffsetScan(C):
  for d in 0 .. 770:
    S ← XOR-decode(C, tableau[d : d+97])
    record IoC(S)
  return max over d, and the offset achieving it

Calibration:
  repeat 1000 times:
    C' ← random shuffle of C's letters
    record OffsetScan(C') # the null replays the WHOLE search
  p ← fraction of null runs ≥ OffsetScan(C)

```

Running the scan with each combination of encoding and out-of-range rule, we found that only the A5 encoding with the passthrough rule provided a significant result. Each encoding/rule combination was tested against 1000 randomly shuffled ciphertexts, the results are shown in Table 2. Raw IoC values are not comparable across runs as the mod 26 and truncation rules concentrate letters regardless of the underlying text. This is the rationale for calibrating each run separately.

Table 2: The full encoding $\times$ out-of-range-rule grid. Each cell is the scan’s max-over-771-offsets IoC, with p from 1000 shuffled ciphertexts.

Encoding	Passthrough	Emit key	Mod 26	Truncate	Reflect
A5	0.0573 ($p=.003$)	0.0503 (.21)	0.0638 (.44)	0.0640 (.11)	0.0595 (.52)
A5 reversed	0.0522 (.07)	0.0507 (.14)	0.0649 (.29)	0.0580 (.61)	0.0627 (.16)
A0	0.0509 (.19)	0.0513 (.11)	0.0586 (.85)	0.0552 (.62)	0.0550 (.75)
KRYPTOS-indexed	0.0492 (.33)	0.0500 (.24)	0.0593 (.52)	0.0535 (.94)	0.0623 (.10)
ITA2	0.0468 (.83)	0.0488 (.44)	0.0601 (.56) ^a	0.0526 (.87) ^a	0.0526 (.57) ^a

^a ITA2’s letter codes are not a contiguous range, so these three rules are not defined for it as written; we realise mod-26 by indexing the 26 letters in code order, and truncate/reflect with a fallback to passthrough when the result is still not a letter.

The scan results indicate a clear signal for the A5/passthrough attempt. Under those rules, the Vigenère tableau has a notable interaction when XORed with K4. We also note that intuitively, A5 is the most natural seeming encoding (it’s just alphabetic indexing) and passthrough is arguably the simplest rule, so a signal in this scan intuitively feels more promising than, for example, the somewhat convoluted ITA2/Reflect would feel.

3.3 Key alignment

The first experiment, summarised in Table 2, showed an interesting pattern exists somewhere among the 771 possible key readings. Next we look at what panel offsets produced significant results in our A5/passthrough scan. This is shown in Table 3.

The best offset is $d = 399$ and to interpret the results, we recall that the engraved tableau’s stream is almost 30-periodic, as each row’s content is the previous row’s content rotated by one letter. This means that other top-scoring candidates, offset from 399 by a multiple of 399, are “echoes” that reuse a very similar but degraded key.

The top-scoring candidate not related to $d = 399$ is the tenth-ranked $d = 439$ and its 0.0481 IoC is just below the null shuffled median of 0.0485. The leading $d = 399$ alignment is also a prominent peak, the gap between it and the second spot is bigger than the gap between #2 and #10. The $d = 399$ candidate is clearly uniquely interesting.

Examining the key alignment that $d = 399$ produces, we make a few observations that are more of an intuitive nature. The first letter of row M’s content (panel position 403, the under-

Table 3: The ten best offsets under A5/passthrough.

#	d	IoC	Δ from 399	Reading
1	399	0.0573	—	best key alignment
2	579	0.0522	+180 = 6×30	row-shift echo
3	519	0.0518	+120 = 4×30	row-shift echo
4	549	0.0518	+150 = 5×30	row-shift echo
5	489	0.0503	+90 = 3×30	row-shift echo
6	609	0.0492	+210 = 7×30	row-shift echo
7	368	0.0490	−31	one whole row up (label included)
8	429	0.0490	+30	row-shift echo
9	398	0.0490	−1	off-by-one neighbour
10	439	0.0481	+40	unrelated; below the null median

row L:FGH
row M: MFGHIJLMNQVWVXZKRYPTOS ABCDEFGHI
row N: N GHIJLMNQVWVXZKRYPTOSABCDEFGHIL
row O: O HIJLMNQVWVXZKRYPTOS ABCDEFGHIJL

Figure 3: The XOR key from the tableau (rows L–O of the panel). The underlined **F** is panel position 399, identified by the scan. content: the key’s origin, aligned to message position 4 — the first letter after **OBKR**. The shaded run is the 61 panel characters the body actually consumes (panel 424–484, keying positions 25–85): it crosses two row boundaries, includes the row-label letters **N** and **O**, and includes row **N**’s extra 32nd character (the underlined **L**), which keys a letter of **BERLIN** (Section 3.5) — and ends exactly at the **S** completing **KRYPTOS** (Section 6.1).

lined **F** of Figure 3) is at K4 ciphertext position 4, right after **OBKR**. This may be coincidental or it may be a point in support of **OBKR** being an indicator group. We believe that only the first two letters **OB** are an indicator pointing to **M**, as discussed in Section 5 but this observation nonetheless felt worth noting.

3.4 K1-K3 as a placebo

Our control was, as mentioned, synthetic texts obtained by randomly shuffling the letters of K4. This control may miss artefacts present in real ciphers so as an additional control, we ran the same key scan on 97-character blocks of K2 and K3, as well as the entire K1. We also ran the algorithm against K1-K3 plaintexts as a sample of real English. None of these runs produced anything of statistical significance, leaving us with a result that doesn’t replicate outside of K4.

3.5 The BERLINLOCK example

We show a brief example, running the proposed mechanism position by position on the **BERLINLOCK** crib (positions 63–73, ciphertext **NYPVTMTZFPK**). At each position the ciphertext letter is XORed with its key letter from the engraved stream, the result is a *stream* letter, which the body’s masking stage links to the plaintext. One position in binary, then all eleven in Table 4:

$$\underbrace{\text{N}}_{01110} \oplus \underbrace{\text{I}}_{01001 \text{ (panel 462)}} = 00111 = \underbrace{\text{G}}_{\text{stream}} \longleftrightarrow \underbrace{\text{B}}_{\text{plain (crib)}}$$

The full de-XORed body stream, with the two crib spans marked, is

PSSVKALSZ ASWTQSQBFSKBZWSMZKCFEBSLHAJ GSPYTTGVKPZ BQSSZSFSSPFT

Table 4: The BERLINLOCK positions under the architecture. The key letters come from panel positions 462–472 (IJLOHIJLMNQ; see Figure 3): they include row N’s extra engraved L (panel 464) and the row-label letter O (panel 465). [†] marks passthrough positions, where $C \oplus K$ falls outside 1–26 and the ciphertext letter passes through unchanged.

Pos	Cipher	Panel pos	Key	$C \oplus K$	Stream	Plain
63	N (14)	462	I (9)	7	G	B
64	Y (25)	463	J (10)	19	S	E
65	P (16)	464	L (12)	28	P [†]	R
66	V (22)	465	O (15)	25	Y	L
67	T (20)	466	H (8)	28	T [†]	I
68	T (20)	467	I (9)	29	T [†]	N
69	M (13)	468	J (10)	7	G	C
70	Z (26)	469	L (12)	22	V	L
71	F (6)	470	M (13)	11	K	O
72	P (16)	471	N (14)	30	P [†]	C
73	K (11)	472	Q (17)	26	Z	K

4 Body and flanks

4.1 The localisation scan

After making the observations detailed in the previous section, the authors spent some effort to extract additional plaintext or structural information from K4 but were unable to make any progress worth reporting.

Progress resumed after thinking through the IoC of 0.0573 that the message has after applying the $d = 399$ XOR key. This statistically significant IoC is well above random but also well below English and the intermediate stream, after reverting XOR, resisted the authors’ attempts.

Then we hypothesised that the XOR key may fit only part of K4, with the rest of the message using no XOR. The IoC of 0.0573 would then likely be a lower-than-actual value, diluted by the part where the XOR should not be applied.

To test this, we generalised the scan (Algorithm box 2): over *every* combination of offset d and contiguous window $[a, a + \ell)$ with $\ell \geq 20$, score the window’s IoC as a z -value against shuffle bands matched to that window length. This is $771 \times 3081 = 2,375,451$ geometries.

Algorithm 2: offset-and-window localisation scan

```

LocalScan(C):
  for d in 0 .. 770:
    S ← XOR-decode(C, tableau[d : d+97])
    for every window [a, a+ℓ), ℓ ≥ 20:
      z ← (IoC(S[a:a+ℓ]) − null mean) / null sd # per-length bands
  return max z and its (d, window)

Calibration:
  repeat 1000 times:
    C' ← random shuffle of C's letters
    record max z of LocalScan(C') # same 2,375,451 geometries
  null max-z ← those 1000 maxima; p ← fraction ≥ real

```

The z -score we use is the IoC of a particular window, standardised against noise of the same length.

$$z = \frac{\text{IoC}_{\text{window}} - \mu_\ell}{\sigma_\ell},$$

where μ_ℓ and σ_ℓ are the mean and standard deviation of IoC over windows of the same length ℓ cut from randomly shuffled K4s passed through the same A5/passthrough decode at random tableau offsets. The controls are thus noise with K4’s letter distribution and no positional information.

For each of 1000 shuffled ciphertexts, only the single best z found across all 2,375,451 of its geometries is kept: that “null max- z ” distribution (median 7.2, maximum 14.6) shows noise routinely manufactures $z \approx 7$ when allowed this much search space.

The LocalScan run once again found the same $d = 399$ key offset as the best geometry, and the top-scoring windows ($z \geq 9.5$) all start at positions 25–26. For every window length ℓ of 44 or more, the best window of that length anywhere in the search has $d = 399$ and overlaps the canonical body [25, 86) this paper proposes. The window boundaries are soft, as one wrong letter barely moves a window’s score, but the rate of loss is itself informative: extending a window costs about 0.07σ per added letter inside the body, and 0.31σ for the first letter beyond the body at either end — position 24 on the left, position 86 on the right.

Section 6.1 discusses the boundaries in more detail, nothing in this section depends on the exact choice. But we will note now that our pick of boundary at 25 is motivated by it being a known word boundary - EAST is at positions 21–24 in K4 and position 25 is a natural break, even if EASTNORTHEAST is a compound bearing.

Table 5: Localisation scan across all geometries. Two calibrations are given: *agnostic* prices the search as a blind expedition (every shuffle gets the identical full geometry search, keeping single best z), the harshest possible accounting; *sequential* replays the two-stage procedure actually used (each shuffle gets the offset scan of Algorithm 1 first, then the window search at its own winning offset).

Quantity	Value	Notes
best geometry	$d = 399$, [26, 93), $z = 9.8$	re-finds the Algorithm-1 alignment
per-length winners	every $\ell \geq 44$ at $d = 399$, over the body	shorter ℓ : noise-level only ($z \leq 7.1$)
body window	[25, 86): IoC 0.0803, $z = 9.4$	adopted here; edges argued in §6.1
agnostic null verdict	max- z median 7.2, 90th 9.0, max 14.6 $p(\text{global}) = 0.044$; $p(\text{body}) = 0.072$	noise, given the same search worst case, if conducted as a blind search
sequential null verdict	max- z median 5.0, max 12.0 $p = 0.007$	both stages replayed per shuffle the procedure as actually run

The two calibrations answer different objections. The agnostic figure prices the localisation as if the search had been run blind, with no OffsetScan before it. This is the harshest accounting, and such a blind search never happened.

The sequential figure tests the composite statistic of the procedure as actually run. Under the null hypothesis, a shuffle’s best whole-stream offset d and its best XOR window have no reason to coincide, so confining the window search to its own first-stage winner drops the median z -score of shuffles from 7.2 to 5.0. For the real K4, if the key engages at a particular alignment, the whole-stream winner and the window host are the same place by design. Note that the sequential p and Algorithm 1’s p are computed on the same data with correlated statistics: each stands alone, and they must not be multiplied.

The canonical body window [25, 86) scores IoC 0.0803, comfortably above English. This suggests the de-XORed interior is output of a substitution with merged letters — the mask of Section 7.1. For this section’s purposes it just means the XOR layer is real and sits on top of something else.

4.2 Flank statistics

Applying the tableau key to the body segment produced a high IoC, but the XOR key has the opposite effect on the flanks. Table 6 shows that both flanks have an IoC below that of

uniformly random text if the XOR is applied to them. This suggests to us that the body and tail zones, which we call the flanks collectively, are enciphered differently.

We briefly considered the possibility that another tableau key is to be used for the flanks, so we ran the tableau key scan over all offsets again, measuring the IoC of only the flanks. No tableau offset produces a statistically significant improvement in the flank IoCs.

Table 6: The three zones when the $d = 399$ key is applied to the entire C4. Reference: random text IoC ≈ 0.0385 , English ≈ 0.0667 .

Zone	Positions	Len	IoC under the XOR
Head	0–24	25	0.0300
Body	25–85	61	0.0803
Tail	86–96	11	0.0364

This emerging body-flank split makes it interesting to compare the raw ciphertext of the zones. For the body and the flank parts, we take its letter-frequency distribution and measure their dissimilarity by the total-variation distance. For each zone, let p_c and q_c be the proportion of letter c in the flank and in the body respectively (so $\sum_c p_c = \sum_c q_c = 1$). Their total-variation distance is

$$TV = \frac{1}{2} \sum_{c=A}^Z |p_c - q_c|.$$

The result is $TV = 0.594$.

As a control, we do 5000 random partitions of the K4 ciphertext into a 61 letter and a 36 letter block, calculating the total variance of such random splits. The median value for random partitions is 0.42 and our calculated $TV = 0.594$ exceeds all but 0.3% of random partitions, giving evidence at $p = 0.003$ that body and flank ciphertext have a different underlying method.

4.3 Entropy measurements

The index of coincidence is one member (the collision, $\alpha = 2$ case) of the Rényi family of entropies. Cross-validating the findings through other entropy calculations is not an independent validation but it does confirm that the statistical anomaly with the identified key is not an artifact of the $\alpha = 2$ parameter in IoC.

An independent verification that this spike is not reproducible from the letter multiset or similar ciphertext alone is the verification against random K4 shuffles and the application of K1-K3 placebos in 3.4.

Table 7: The $d = 399$ key anomaly across other entropy measurements. Calculated for the de-XORed body stream, All four place offset 399 first of 771. Entropies in bits.

Statistic	$d=399$	English (reference)	Offsets (mean $\pm$ sd)	Rank
Index of coincidence	0.0803	0.0655	0.040 ± 0.005	1/771
Rényi-2 entropy	3.39	3.93	4.16 ± 0.13	1/771
Shannon entropy	3.82	4.18	4.33 ± 0.11	1/771
Min-entropy	2.12	2.98	3.36 ± 0.23	1/771

4.4 K4 zones with the key

Figure 4 lays the whole arrangement out, position by position. The key stream read from the tableau at $d = 399$ runs alongside the entire message, with the XOR operation only applied in

The boxed **L** is the extra 32nd character in the tableau's N row.

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performing an XOR on their values also points to M.

$$O \oplus B = 15 \oplus 2 = 13 = \text{'M'}$$

We speculate that the letters **OB** are such an intentional hint. This would make **OB** an *indicator*, a common technique (used by Enigma among others) where part of the transmitted cipher, usually at the beginning, is information about the key.

Making **OB** an indicator would be very challenging if the XOR applied to the entire K4 ciphertext. If however the XOR only applies to the body zone, as we propose, it would be much easier to encrypt the flanks in a way that the first two letters are **OB**, indeed it would be trivial if the flanks use a substitution of some kind.

6 Narrowing the zones

In earlier sections we located a tableau key and a body window to which it applies by search. In this section we look at some alternative hypotheses and we also examine how precisely we know the boundaries of the body and flank zones.

If the first two letters **OB** are meant to be an indicator for the XOR key, there is an obviously simpler cipher design. Leave the indicator letters exactly as they are in the plaintext and then apply masking and XOR to the rest of the message. So the plaintext begins with **OB** (there are certainly many English words starting so) and the remaining 95 characters are all masked by the unknown procedure and then XORed with the key.

We cannot definitively rule out such a construction method, but the statistics given in 4.2 are against it. If K4 is indeed split into a body and flanks, we want to narrow down the exact boundaries.

6.1 Head–body boundary

Our localisation scan cannot confidently indicate the zone boundaries. Moving a window by one position back, while keeping the length, will at worst exchange one correct position for noise, barely affecting the IoC. The per-length winners in Table 5 have a boundary at 25 or 26. So the scan leaves us with a ± 1 resolution.

No per-length winner in the scans started at 24 or earlier. Given the choice between position 25 or 26 as the body start, we pick 25 mainly due to a plaintext boundary, with some support from Bayesian modeling.

Our global offset scan and the localisation scan operated on the K4 ciphertext and the sculpture’s tableau, without any knowledge of the cribs. The known plaintext **EAST** is at positions 21–24 and **NORTHEAST** starts at 25. If the encryption method changes at the boundary, it seems far more natural for the boundary to be at position 25, splitting **EAST|NORTHEAST** in a way that aligns with the English words of the plaintext.

We also ran an ABC (Approximate Bayesian Computation) model generating 3 million synthetic K4s (using the known cribs, the remaining characters filled with English words) and applying a XOR key from a random panel offset to a random window. The windows were given start locations at positions 21 – 33 and a length of 50 – 70. We scored these synthetic K4s on sixteen dimensions, of which eight contain no IoC measurement (five top letter frequencies, XOR passthrough fraction, autocorrelation, flank-body TVD) and eight were different IoC computations ¹.

Comparing the synthetic K4s against the real one, the ABC posterior scored position 25 as the most likely, in the full 16-dimension evaluation and the 8-dimension IoC-free evaluation.

¹Over whole stream; chunks at [0, 16), [16, 32), [32, 49), [49, 65), [65, 81), [81, 97); bigrams over whole stream

The results in 8 show that positions 25 and 26 are favoured. While 25 scores marginally higher, its lead over 26 is not conclusive.

Table 8: ABC posterior over the body-window start from 3,000,000 synthetic K4s, under the full 16-dimensional signature and the no-IoC 8-dimensional subset. Both peak at position 25. Prior: uniform over 21–33.

Window start	Full (16-dim)		No-IoC (8-dim)	
	n	%	n	%
21	301	2.9	315	3.6
22	510	4.9	470	5.4
23	690	6.6	538	6.2
24	810	7.8	661	7.6
25	1154	11.1	871	10.0
26	1087	10.4	867	10.0
27	1075	10.3	842	9.7
28	991	9.5	799	9.2
29	915	8.8	786	9.0
30	887	8.5	735	8.4
31	766	7.4	716	8.2
32	631	6.1	588	6.8
33	601	5.8	520	6.0
accepted	10,418		8,708	

6.2 Body–tail boundary

The end of the window, the body–tail boundary, is a softer one. Statistically, the tail could begin at positions 83 – 86 with little to distinguish these picks. z , the window’s IoC standardised against length-matched shuffles, remains high and flat across the 83 – 86 range, with Δz being at noise levels until the tail boundary is moved to position 87.

Table 9: Body/tail boundary: localisation z versus the candidate tail-start position, the body window is [25, tail start). Bands carry $\pm 0.06\sigma$ Monte Carlo noise.

Tail start	IoC	z	Δz
83	0.0835	9.61	—
84	0.0824	9.53	−0.09
85	0.0814	9.48	−0.05
86	0.0803	9.37	−0.10
87	0.0783	8.97	−0.40

Our main argument for setting the tail start at 86 arises from looking at the XOR key readout. At positions 79 – 85, the key is KRYPTOS (from row O of the tableau). For an intentionally designed encryption scheme, it seems more natural to set the boundary after KRYPTOS.

Under the body window [25, 86), the running XOR key for the body zone starts with ABC and ends with KRYPTOS, an attractive pattern that treat as the canonical boundary for our proposed architecture.

7 Decryption possibilities

Under the proposed K4 architecture, we have three zones with different encryption methods. The individual flanks, head and tail, are short and we assume that both flanks use the same encryption method, which makes K4 a two-cipher message.

In this section we examine what the three zone split and the XOR key tell us about the possibilities of retrieving plaintext. The authors have not given this question an exhaustive examination but the initial findings do not show any clear path towards decrypting K4 assuming the paper’s architecture.

7.1 Body’s stream cipher

After undoing the XOR operation on the body, we get the intermediate stream, which is itself the result of an unknown masking cipher and contains two cribs.

Unfortunately, the obtained stream does not offer many pointers. It has many **S** characters, indicating that the masking cipher merges letters, and the known cribs do not fit any simple substitution model. We see that four different plain text letters (**E**, **O**, **R**, **S**) become **S** in the stream, with other many-to-one mappings also existing.

Cipher	QQPRNGKSSOTWTQSQSSEKZZWATJKLUDIAWINFBNYPVTMTZFPKWGDKZXTJCDIG
Stream	PSSVKALSZASWTQSGQBFSKBZWSMZKCFEKBLSLHAJGSPYTTGVKPBZBQSSZSFSSPFT
Plain	NORTHEAST.....BERLINCLOCK.....

Figure 5: The body segment (K4 positions 25–85, 61 letters).

The observed patterns do allow for two substitution methods at the masking layer that the authors identified.

There could be two substitution tables, call them T_0 and T_1 , applied in a period-7 pattern ($T_0, T_1, T_0, T_0, T_0, T_1, T_1$). Each table maps one or two plaintext letters to a stream letter. Under this two-table reading, the plaintext appears unrecoverable. This is also just one fit among several, the cribs are also consistent with a longer period or more tables.

The same observations could be explained by an occurrence counting system, where each plaintext letter has a list of (likely two) stream letters, with the first occurrence of the plaintext letter being substituted for the first letter on the list, etc. Under this scheme there is likewise insufficient information to recover the plaintext.

Regardless of the exact scheme, the problem is evident in the **NORTHEAST** crib alone. The stream contains three letters **S** that map to different plaintext letters, which is a strong alphabet collapse. Such an encryption scheme is possible but it is likely not reversible without knowledge of the exact process.

A particularly interesting segment is at the end of the body stream. We see **SSZSFSSPFT** ending the stream, a segment very heavy on the letter **S** and significantly contributing to the measured IoC. In the **S**-heavy body, 12 of 14 occurrences are from XOR, not passthrough, and the XOR of two uniformly-distributed letters favors small output values so we note this as an interesting pattern.

We emphasise that this is not a definitive conclusion as more investigation of the stream as ciphertext can be conducted.

7.2 The flank cipher

For our investigation of the flanks, we treated them both as a continuous span of ciphertext encrypted by one unknown method. The flanks contain one crib, **EAST**.

The authors hit a dead end quite rapidly with the flanks. A monoalphabetic substitution cipher is possible but places difficult constraints on the plaintext (**T** may not appear outside

of EAST, only one more appearance of E) and we were not able to obtain even speculatively meaningful tilings. A polyalphabetic substitution for the flanks is minimally constraining and admits almost any text.

A Vigenère search similar to K1-K2 turned up nothing for the flanks, and a brute force search of ACA Gromark in the standard and KRYPTOS-keyed alphabets likewise produced no results. Some other classical ciphers are excluded due to the flank structure, like Playfair being impossible due to the BB digraph.

8 Conclusion

We have proposed a three-zone structure for K4, where the middle zone is to be decrypted by XORing it with a running key from the tableau panel.

The flanks, the head and tail zones, use a different mechanism.

The architecture, if correct, would explain the meaning of the raised letters in DYHR and the reason for the extra L in one of the tableau's rows.

At the present time, we do not see a clear path from the architecture to a plaintext recovery and that remains an avenue for future work.

References

- [1] C. Hörenberg, *The Kryptos Sculpture: K4 analysis*, web publication, <https://kryptos.hoerenberg.com>. Origin of the tableau-XOR model and the row-M alignment calibrated in this work.